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Organism response to habitat patchiness: species and habitat-dependent recruitment of decapod crustaceans

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Abstract

Habitat fragmentation from natural or anthropogenic causes is a common phenomenon in shallow water marine habitats such as seagrass beds and oyster reefs throughout the world. Thus, habitat- and scale-dependent information on how organisms perceive and respond to changes in the landscape is critical to efforts aimed at predicting the effects of habitat change on population abundance. In estuarine systems, seagrass is considered one of the predominant nursery habitats for numerous ecologically and commercially important species; however, recent information suggests that alternative habitats such as oyster reefs may also be important sites for settlement and refuge. We used artificial seagrass- and shell-filled plastic trays placed on unstructured seafloor in Back Sound, North Carolina, USA to examine the interactive effects of Experiment, Patch size and Habitat type upon recruitment success of four species of decapod crustaceans: the grass shrimps, *Palaemonetes intermedius* Holthuis, *P. pugio* Holthuis and *P. vulgaris* (Say), and the blue crab, *Callinectes sapidus* Rathbun. Regardless of habitat type, all species of grass shrimp and J1 blue crab responded to habitat patchiness at scales as small as 0.25 to 4 m^{-2}, whereas larger J2–J4 blue crab did not. Mean densities of all three species of grass shrimp responded in a similar manner to patch size, with significantly higher densities in the smallest patch size (0.25 m^{-2}) compared to larger patches (0.5 to 4 m^{-2}). Conversely, mean densities of J1 blue crabs were significantly higher in the largest patch size (4 m^{-2}) compared to smaller patches (0.25 to 1 m^{-2}). There was a negative and significant relationship between the density of grass shrimp predators and density of J1 prey, and no relationship between potential fish predators and J1–J4 prey. Thus, patterns observed for J1 may be dependent upon predators and not habitat patch size. Both species demonstrated different responses to habitat type depending upon relative densities and body size. Mean densities of *P. intermedius* and *P. vulgaris* were higher in seagrass than oyster shell during Experiment 1 (May) when shrimp were relatively abundant, but did not differ between habitat types during Experiment 2 (October), when densities were reduced 10-fold. Conversely, densities of *P. pugio* did not differ between seagrass and oyster shell. Mean densities of J1 blue crab were similar between seagrass and oyster shell, but higher in seagrass than oyster shell for later juvenile stages (J2–J4). Thus, blue crabs may rely equally on seagrass and oyster reefs as an initial

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settlement and refuge habitat for early juveniles. Our study demonstrates that an organism's response to habitat patchiness is species-specific, and that for a given species, the response is further modified by animal density and body size. The patterns observed in this study further highlight the importance of scale-dependent responses by mobile organisms to complex benthic habitats. © 1998 Elsevier Science B.V.

Keywords: Callinectes sapidus; *Crassostrea virginica*; Fragmentation; Grass shrimp; Habitat selection; Palaemonidae; Patchiness; Recruitment; SLOSS; *Zostera marina*

1. Introduction

The spatial scaling of organism–structure relationships deserves greater emphasis in ecological studies because organisms differ greatly in those morphological, physiological and behavioral characteristics that influence the way in which habitat patchiness is perceived (Pickett and White, 1985; Frid and Townsend, 1989; Hart and Horwitz, 1991). There are no strict conventions regarding the spatial scale over which an organism's response to habitat patchiness should be measured (Hart and Horwitz, 1991). The smallest scale at which an organism responds to patch structure by differentiating among patches is its 'grain' (Kotliar and Weins, 1990). At smaller scales, the organism functionally perceives its environment as homogeneous and does not respond to any structure that might exist (Kolasa, 1989). The lower limit to grain is set by the physiological and perceptual abilities of the organism, but behaviour may constrain grain to somewhat larger spatial scales (Kotliar and Weins, 1990; Hart and Horwitz, 1991). 'Extent' is the largest scale of heterogeneity to which an organism responds (Kotliar and Weins, 1990). This upper limit is determined by the lifetime home-range of the individual. The particular scales of grain and extent will vary with the types of organisms studied. Specifications of grain and extent are essential because they demarcate the context of investigations of patchiness at a particular scale within a hierarchy of patches, and because they provide a framework based on scale in which comparisons of relative patchiness between systems may be standardized (Kotliar and Weins, 1990).

Seagrass is often the dominant subtidal vegetation in nearshore coastal marine environments, and supports large standing crops and high species diversity of organisms (e.g., Lewis and Stoner, 1983; Sogard and Able, 1991; Orth, 1992 and references therein). High macrofaunal biomass combined with spatial patterning ranging from monospecific beds that extend over several kilometers, to fragmented patches <0.25 m^{-2}, make seagrass an ideal system for assessing the effects of habitat patchiness on animal abundance (Robbins and Bell, 1994; Bell et al., 1995; Fonseca, 1996). Seagrass beds exhibit natural expansion, fragmentation and contraction due to seasonal growth, die-off, and storm events (e.g., Duarte and Sand-Jensen, 1990; Fonseca, 1992). Seagrass meadows may also become fragmented through the feeding activities of epibenthic predators (e.g., crabs and rays) and herbivores (e.g., sea urchins), and through anthropogenic impacts such as damage from boat propellers and dredging (e.g., Peterson et al., 1987; Valentine and Heck, 1991; Fonseca, 1992).

Subtidal and intertidal oyster reefs also possess a suite of spatial and ecological characteristics that make them amenable to assessing the effects of habitat patchiness on animal abundances (Eggleston, in press). For example, oyster reefs range in size from small, fragmented intertidal reefs of less than 1 m^{-2}, to continuous subtidal reefs that extend over 1 km. The shape and morphology of these reefs also varies considerably, ranging from small, cup-shaped oysters that form low relief subtidal reefs (Mann et al., 1991), to thin-shelled, elongated oysters that make up narrow strands of intertidal reefs (Kennedy and Sanford, in press). Reef expansion occurs through oyster growth and enhanced larval settlement on large conspecifics (Haven et al., 1978; Kennedy and Briesch, 1981), whereas reef reduction is due to habitat destruction from fishing and dredging practices, and the death of reef assemblages due to disease, physical disturbances, predation and sedimentation events (Eggleston, 1988; Armstrong et al., 1989; Osman et al., 1989; Mann et al., 1991; Rothschild et al., 1994). In many areas, oyster reefs can then be placed within a larger coastal landscape containing a mosaic of seagrass, salt marsh and unstructured soft-bottom habitats.

1.1. Effects of habitat fragmentation in marine versus terrestrial systems

A frequently debated prediction in conservation biology is whether a single large reserve will contain more or less species (or density of a given species) than several small reserves of the same total area; the so-called SLOSS (Single Large Or Several Small) debate. The general prediction from terrestrial systems is arguably increased species abundance and diversity with large habitat areas (small perimeter:area ratio), and decreased abundance and diversity with increasing detrimental edge effects (i.e., increased perimeter:area ratio, see review by Saunders et al., 1991; Robinson et al., 1995; Rosenzweig, 1995). Conversely, recent studies in seagrass systems suggest that many small seagrass patches may increase the overall probability of encounter by larvae or other immigrants, thereby increasing overall colonization of the patch, compared to larger patches (e.g., Bell et al., 1987; Sogard, 1989; Worthington et al., 1992; McNeill and Fairweather, 1993; but see Bell et al., 1995, for an example of disproportionately high algal accumulation rates on large versus small patches). These observations prompted one research team to propose a conservation strategy for seagrass associates that attempts to preserve discrete seagrass beds within a larger area, rather than preserve one large contiguous bed (McNeill and Fairweather, 1993). A similar management strategy was proposed for conserving breeding waterbird populations in Florida Bay, USA (Erwin et al., 1995). Increased recruitment of barnacles into small versus large habitat patches has also been observed in rocky shore habitats (Paine and Levin, 1981; Keough, 1984; Sousa, 1984; but see Kim and DeWreede, 1996 for an example of high barnacle densities in intermediate and large size patches). The hypothesized mechanism for the scale-dependent patterns observed in rocky shore habitats was similar to that posed for seagrass; increased probability of interception of larvae by the patch edge in small versus large patches (Paine and Levin, 1981; Sousa, 1984). Differences in the response of organisms in terrestrial versus marine systems to habitat patchiness, if they prove to be general, may be due to greater dispersal limitation in terrestrial versus marine systems (e.g., Robbins and Bell, 1994), and the increased negative impacts that

physical fluxes (e.g., radiation, wind, water) have across smaller patches in terrestrial systems (Saunders et al., 1991).

For postlarval and early juvenile stages of grass shrimp and blue crabs, the spacing of seagrass and oyster shell among unstructured, soft-bottom habitat is relatively large (1–100 m) compared to their body size (0.5–10 mm). Thus, mosaics of sediment, seagrass and oyster reefs may directly or indirectly impact populations through a variety of mechanisms such as alteration of predator (e.g., shrimp, fish) distribution, abundance and foraging behaviour (e.g., Leber, 1985; Main, 1987; Bell and Hicks, 1991; Irlandi, 1994; Irlandi et al., 1995), modifications of water flow which may passively entrain or deposit settlement stages (e.g., Eckman, 1983; Bell et al., 1995 for an example with passive accumulation of algae), and changes in animal behaviour (review by Heck and Crowder, 1991). Thus, recruitment success of marine benthic species in complex habitats may be scale-dependent.

Information on the interdependence of shallow estuarine habitats such as seagrass and oyster reef habitats, as well as their relative importance as settlement and refuge sites, is critical for understanding population dynamics of numerous commercially and ecologically important estuarine-dependent species such as the grass shrimp (*Palaemonetes* sp.) and blue crab (*Callinectes sapidus* Rathbun). This information is particularly important in areas lacking seagrass (e.g., southern North Carolina southward through Georgia, USA), where oyster reefs may represent the only structural refuge available. The general lack of information on how grass shrimp and blue crabs respond to habitat patchiness, combined with recent findings highlighting the relative importance of alternative habitats such as oyster reefs to grass shrimp and blue crab abundance (Coen et al., in press; Posey et al., in press), prompted us to address two questions. (1) What are the interactive effects of patch size and habitat type upon shrimp and crab densities? (2) What is the relationship between densities of early juvenile blue crabs and their potential predators?

2. Materials and methods

2.1. Study area

Two 14-day experiments were conducted on May 20–June 2, 1995 (Experiment 1), and October 13–27, 1995 (Experiment 2) at a shallow subtidal site within Middle Marsh, in Back Sound, North Carolina, USA (Fig. 1). Water depth at the site ranged from 0.1–0.4 m at low tide to 1.2–1.5 m at high tide. Water temperature and salinity during the two experiments averaged 25°C and 35 ppt (Experiment 1), and 20.5°C and 32 ppt (Experiment 2), respectively. Salinity is high all year because of relatively low riverine input and high tidal flushing through Beaufort Inlet (Fig. 1). The experiments were conducted on an unstructured bottom within a small cove surrounded by *Spartina alterniflora* Loisel marshes. The closest seagrass bed or oyster reef was located approximately 20 m away. Sediments were primarily muddy–sand, with 5.6% silt–clay and 89.9% very fine sands (F. Micheli, unpubl. data). The Middle Marsh site typifies the Albemarle–Pamlico estuarine system and is characterized by extensive seagrass (pri-

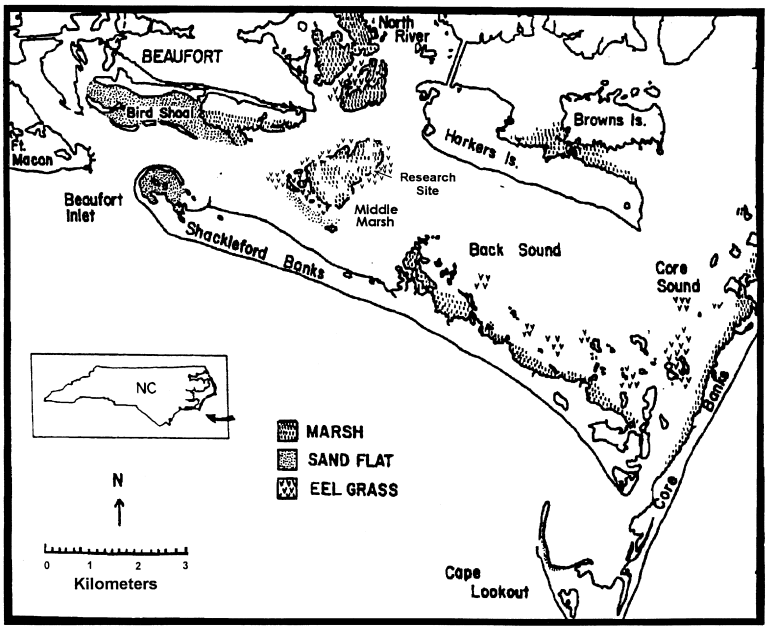

Fig. 1. Study site in Middle Marsh, Back Sound, North Carolina, USA.

marily *Zostera marina* and *Halodule wrightii*), marsh creek and inter- and subtidal oyster (*Crassostrea virginica*) habitats.

2.2. Artificial settlement plots

The most direct means of testing the effects of habitat fragmentation on biota is by comparing the density of a given species (or number of species per unit area), measured from samples of equal size, from increasingly larger habitat patches or segments (e.g., Coleman et al., 1982; Haila, 1983; Kelly et al., 1989; Hart and Horwitz, 1991; McGarigal and McComb, 1995; and references therein). In this study, we assessed the interactive effects of patch size and habitat type on densities of grass shrimp (*Palaemonetes pugio Holthuis*, *P. vulgaris Say*, *P. intermedius Holthuis*) and early juvenile blue crabs with artificial seagrass and trays filled with dead oyster shell.

Artificial seagrass plots were constructed of green plastic ribbon (Equality Specialty, Chicago, Illinois, USA) tied to 0.25 m^{-2} squares of black plastic (Vexar) with a mesh size of 25 mm. Each shoot consisted of two blades, 30 cm in height and 5 mm wide, with a total shoot density of 2704 m^{-2}. Natural densities of patchily distributed seagrass in this area during October range from 2651 to 2917 m^{-2} (Irlandi, 1994). Each seagrass plot was supported by a 0.25-m^2 plastic frame (50 cm length $\times$ 50 cm width $\times$ 5 cm height) lined with 500-μm mesh. Seagrass plots were anchored to the bottom by placing a 'J-shaped' metal stake on each corner of a tray and pushing it into the sediment.

Artificial oyster plots were constructed by filling a 500-μm mesh-lined 0.25-m^2

plastic tray with 19 liters of dead oyster shell. The substratum surface in these trays stood 5 cm above the surrounding sediment surface. Although the use of dead oyster shells does not represent the complex architecture of natural intertidal reefs (e.g., Wells, 1961; Coen et al., in press), it does mimic low vertical profile subtidal reefs found throughout the East and Gulf coasts of North America (Mann et al., 1991; Rothschild et al., 1994).

2.3. Experimental procedure and design

The experiments were timed to correspond with peak abundances of grass shrimp in North Carolina during the late spring/early summer (Experiment 1), and a minor peak in the fall (Experiment 2) (Rozas and Hackney, 1984). The second experiment was also timed to coincide with peak ingress of blue crab megalopae into Back Sound. Seagrass and oyster shell settlement trays were placed on the bottom during the end of the full moon, low tide in May (Experiment 1) and October (Experiment 2) 1995, then retrieved 14 days later at the end of the new moon, low tide. During the second experiment (October 24–26), a moderately large pulse of blue crab megalopae was recorded nearby at the Duke University Marine Laboratory just north of Bird Shoal (Fig. 1) (R. Forward, unpubl. data). The 14-day experimental period was chosen because it encompassed a new moon, spring tide, which would presumably enhance recruitment because of the relatively large tidal flux of water into the system, and because it was the shortest period that would permit deployment and retrieval of the habitat plots during low tides. We had no a priori evidence to suggest that one experimental period should have been chosen over another (e.g., 14-day versus 4 months).

We assessed the direct and interactive effects of patch size and habitat type on recruitment of grass shrimp and blue crabs with four similarly shaped patches of varying size (0.25, 0.5, 1, 4 m^{-2}) and two habitat types (oyster shell, seagrass) (Fig. 2). Our choice of patch sizes was based on the size distribution of natural patches in the area and ease of deployment. For example, the size distribution of natural seagrass patches surrounding Middle Marsh is skewed greatly towards patches that are $< 4\ m^{-2}$ (Irlandi, 1994; Eggleston, unpubl. data). A similar patch size distribution has been identified in *Z. marina* beds in Denmark (Olesen and Sand-Jensen, 1994).

In terms of assessing how the density of an organism changes as a function of a single large versus several small patches of equivalent area (the SLOSS debate), it is instructive to know how the ratio of perimeter to area varies with patch size. When we standardized our patch sizes to a total area of 4 m^{-2} (the largest patch size used in this experiment), the 0.25, 0.5 and 1 m^{-2} patches contained, respectively, four, three and two times as much perimeter as the 4 m^{-2} patch. Thus, in terms of patch colonization, we predicted that the higher perimeter associated with smaller patches should result in higher encounter rates with these patches than larger ones, leading to higher densities of animals in small versus large patches.

The experimental design for the effects of patch size and habitat type was orthogonal with replicate treatments (Habitat type $\times$ Patch size; $N = 3$) randomly assigned to positions along four 70-m transect lines (Fig. 2). To minimize disturbance of treatments

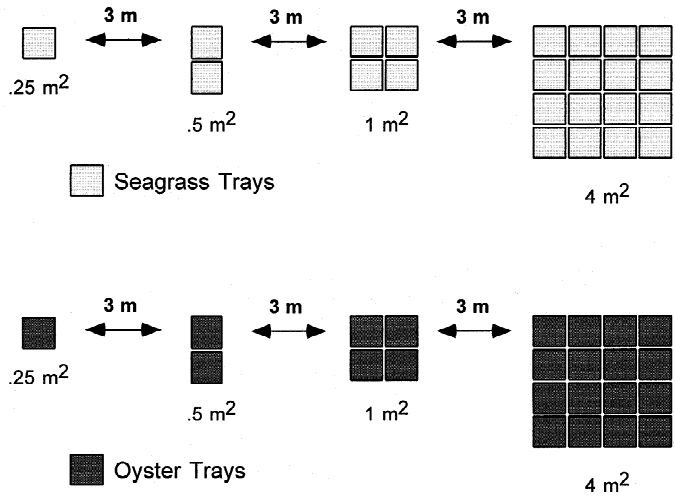

Fig. 2. Schematic of experimental design. All replicates ($N = 3$) of each treatment were randomly interspersed along three 70-m transect lines.

during retrieval and to help insure statistical independence, all treatments were located 3 m apart. When patch sizes were greater than 0.25 m^{-2}, we subsampled the patch by randomly selecting a single tray for retrieval. Only trays along the edges of the 4 m^{-2} patches were randomly sampled. Trays were retrieved by quickly placing a 0.25 m^2 box sieve with 500-μm mesh around the edges of a tray, and lifting the tray out of the water. Retrieval of trays was usually conducted when water depths were approximately 0.1 m. This retrieval approach was presumably effective at retrieving grass shrimp and early juvenile blue crabs since large, highly mobile blue crabs (<16 cm carapace width; CW) and pinfish, *Lagodon rhomboides* (Linnaeus) (<12 cm total length; TL) were effectively captured with this technique. Upon retrieval, individual trays were placed into plastic bags with an identification tag; the contents of a tray were then sieved with a 500-μm screen and organisms preserved in 10% formaldehyde solution for later enumeration.

All shrimp and crabs were identified to species in the laboratory. Four families of shrimp were represented: Palaemonidae, Alpheidae, Penaeidae and Hippolytidae. Because of their high abundance, only palaemonids were examined with respect to patch characteristics and habitat type. Moreover, palaemonids are considered potential predators of blue crab megalopae and early juveniles (Leber, 1985; Olmi and Lipcius, 1991). The primary species of hippolytid shrimp that we found, *Tozeuma carolinense*, was not considered a potential predator based on its slender body shape and small size (Main, 1987). The size of palaemonid shrimp (mm carapace length; CL) was determined by measuring the distance between the eye hood and end of the carapace. Only palaemonid shrimp >5.1 mm CL were considered as potential predators on megalopae and early juveniles (Olmi and Lipcius, 1991). We did not consider other potential

predators such as Penaeid shrimp, fish and xanthid crabs due to their very low abundances.

Early juvenile *Callinectes sapidus* were distinguished from *C. similis* using the criteria established by Stuck and Perry (1982). Blue crab instars were classified according to Newcombe et al. (1949) as J1 (2.2–3.0 mm CW), J2 (3.1–4.2 mm CW), J3 (4.3–5.9 mm CW), and J4 (6.0–7.4 mm CW). Time-to-metamorphosis from the megalopal stage to J1 in the blue crab ranges from hours to 4 days (Wolcott and De-Vries, 1994). After settlement, early juvenile blue crabs typically reside in seagrass habitats until they reach a stage between the fifth and seventh instar (8–11 mm CW); after which they migrate out of seagrass and into non-vegetated habitats (Orth and van Montfrans, 1990; Heck and Coen, 1995; Pile et al., 1996). The intermolt period from J1 to J4 is approximately 10–14 days (Pile et al., 1996). Although we initially assumed that the majority of J1–J4 collected in our artificial substrata had settled there, recent findings suggest that at least some of these crabs may have immigrated into our plots after settlement elsewhere. For example, an accidental release of stained (neutral-red) J1 from growth enclosures in the lower York River, USA were captured in plankton nets three weeks later (as J2–J3 stages) approximately 10 km upstream of the point of release (Perkins et al., 1996). Thus, densities of early juvenile crab in our settlement trays probably reflected a combination of settlement by megalopae, immigration and post-settlement predation, and cannibalism. Therefore, our definition of recruitment in this study included the following processes: settlement, immigration and post-settlement emigration and mortality.

2.4. Data analyses

We examined the independent and interactive effects of experiment, patch size and habitat type on shrimp and blue crab abundances with various univariate ANOVA models. For grass shrimp, the density of three species of palaemonid shrimp (*Palaemonetes intermedius*, *P. vulgaris*, *P. pugio*) per 0.25 m^{-2} served as separate response variables in a three-way ANOVA model with Experiment (Experiment 1: May; Experiment 2: October), patch size (0.25, 0.5, 1, 4 m^{-2}), and habitat type (seagrass, oyster shell) as factors. For blue crabs, the density of J1 or J2–J4 per 0.25 m^{-2} served as separate response variables with patch size and habitat type as factors. These two response variables (J1, J2–J4) were chosen because no megalopae were collected in the trays and the densities of J2, J3 and J4 were too low to analyze separately. Thus, densities of J2–J4 were pooled for statistical analyses. We used log of $(x+1)$ transformations when variances were heteroscedastic (as measured by Cochran's C-test). Differences between means were revealed with either a protected, lower-level ANOVA (with 2-levels of a treatment), or a Tukey's multiple comparison test ($>$2-levels of a treatment).

The causal relationship between densities of early juvenile blue crabs and the abundance of potential shrimp predators or blue crab cannibals (J2–J4) was examined with linear least-squares regression models. Residuals from these analyses were examined visually to detect departure from randomness.

3. Results

3.1. Effects of patch size, habitat type and experiment on grass shrimp density

During May 1995, a total of 378 *Palaemonetes* sp. were collected from our experimental plots during the 14-day experiment. Three species of palaemonids were found: *Palaemonetes intermedius*, *P. vulgaris* and *P. pugio*. These shrimps averaged 4.5 mm (ranging from 2.2–6.4 mm CL; SD = 1.1), which is indicative of primarily mature, adult stages (Anderson, 1985). In October 1995, a total of only 90 *Palaemonetes sp*. were collected, at an average size of 4.9 mm (ranging from 2.4–6.1 mm CL; SD = 1.0). Abundance patterns of all three species followed a similar trend with respect to patch size, with densities increasing with decreasing patch size. Grass shrimp density was generally higher in seagrass than oyster shell when shrimp were abundant in May (except for *P. pugio*), whereas there was no difference in density between habitat types when abundances of grass shrimps was low in October (Fig. 3).

The density of *Palaemonetes intermedius* varied according to Experiment, Habitat type and Patch size (Fig. 3, Table 1). A significant Habitat type × Experiment interaction effect (Table 1) precluded contrasts across the means of the Habitat type and Experiment main effects (Underwood, 1981). The density of *P. intermedius* increased with decreasing patch size in both oyster shell and seagrass (Fig. 2), and was significantly higher in the smallest patch size (0.25 m^{-2}) than in larger patch sizes, irrespective of Experiment (Fig. 3; Tukey's multiple comparisons test). The Experiment × Habitat type interaction effect was due to significantly higher shrimp densities in seagrass than in oyster shell during Experiment 1 in May, and no difference between habitat types during Experiment 2 in October (Fig. 3; lower-level ANOVAs). None of the other interaction effects were significant (Table 1).

The density of *Palaemonetes vulgaris* varied in a manner similar to *P. intermedius*, with significant Experiment and Patch size main effects, and a significant Experiment by Habitat type interaction effect (Fig. 3 Table 2). The mean density of *P. vulgaris* was significantly higher in the smallest patch size (0.25 m^{-2}) than in larger patch sizes (Fig. 3; Tukey's multiple comparison test). The Experiment × Habitat type interaction effect was due to significantly higher mean densities of *P. vulgaris* in seagrass versus oyster shell when densities were high in May, and no difference in shrimp density between habitat types when densities were low in October (Fig. 3; lower-level ANOVA). None of the other interaction effects were significant (Table 2).

Although densities of *Palaemonetes pugio* were significantly higher in the smallest patch size than larger patch sizes (Fig. 3 Table 3; Tukey's multiple comparison test), similar to the pattern observed for *P. intermedius* and *P. vulgaris*, there was no Experiment by Habitat type interaction effect (Table 3). Rather, there was no difference in mean densities of *P. pugio* between seagrass and oyster shell, irrespective of Experiment (Fig. 3 Table 3). Mean densities of *P. pugio* were higher in May than October (Fig. 3 Table 3; lower-level ANOVA), similar to the trend observed for *P. intermedius* and *P. vulgaris* (Fig. 3).

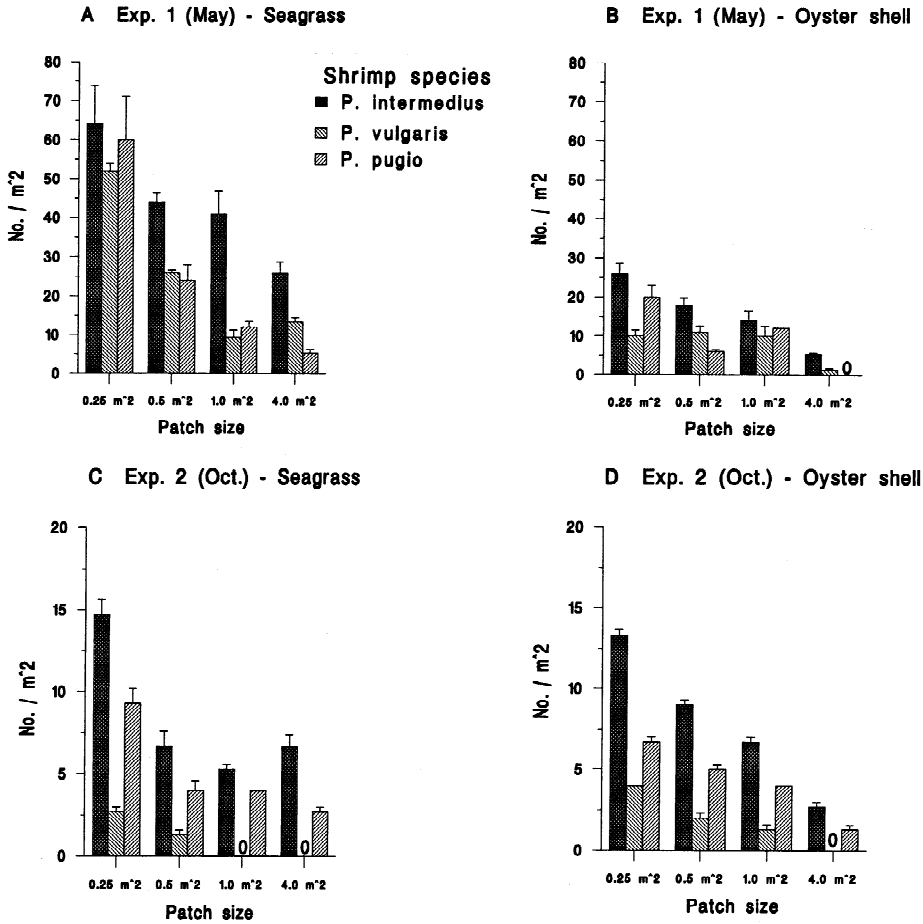

Fig. 3. Mean density (+ 1 SE) of grass shrimp (*Palaemonetes pugio*, *P. vulgaris*, *P. interuptus*) in four patch sizes of artificial seagrass and oyster-shell filled trays placed on an unstructured bottom during Experiment 1 (May) and Experiment 2 (October), 1995. Note different scales on *y*-axes between figures A, B and C, D. $N = 3$. See Tables 1–3 for significance levels.

3.2. Effects of patch size and habitat type on blue crab density

A total of 209 J1–J4 stages of *Callinectes sapidus* were collected from settlement trays during the 14-day experiment in October. Of these, 79% were J1, 6% were J2, 13% were J3 and only 2% were J4. No megalopae were collected. The density of J1 varied according to Patch size, but not Habitat type (two-way ANOVA; Patch size: $F = 5.58$, $df = 3{,}16$, $P = 0.008$; Habitat type: $F = 0.07$, $df = 1{,}16$, $P = 0.80$). The interaction effect was not significant ($F = 1.41$, $df = 3{,}16$, $P = 0.28$). The density of J1 was significantly higher in the largest patch size (4 m^{-2}) compared to smaller patches (Tukey's multiple comparison test; Fig. 4a). Conversely, the log-transformed density of a later stage *C.*

Table 1
Effects of experiment (Experiment 1: May vs. Experiment 2: October 1995), habitat type (artificial seagrass versus oyster shell), and patch size (0.25, 0.5, 1, 4 m^{-2}) on the mean density of *Palaemonetes intermedius*

Three-way ANOVA (model I).

Source of variation	*df*	MS	*F*
Experiment (A)	1	17.01	401.16[b]
Habitat type (B)	1	0.45	10.59[a]
Patch size (C)	3	0.21	5.01[a]
A × B	1	0.56	13.18[b]
A × C	3	0.01	0.07 NS
B × C	3	0.06	1.34 NS
A × B × C	3	0.01	0.17 NS
Error	30	0.04	

Shrimp density was log-transformed $(x + 1)$ to standardize variances. $N = 3$.
NS > 0.05; [a] $P < 0.003$; [b] $P < 0.001$.

Table 2
Effects of Experiment (Experiment 1: May vs. Experiment 2: October 1995), habitat type (artificial seagrass versus oyster shell), and patch size (0.25, 0.5, 1, 4 m^{-2}) on the mean density of *Palaemonetes vulgaris*

Three-way ANOVA (model I)

Source of variation	*df*	MS	*F*
Experiment (A)	1	1.82	28.98[b]
Habitat type (B)	1	0.19	3.05 NS
Patch size (C)	3	0.27	4.23[a]
A × B	1	0.55	8.78[b]
A × C	3	0.04	0.71 NS
B × C	3	0.05	0.78 NS
A × B × C	3	0.05	0.75 NS
Error	30	0.06	

Shrimp density was log-transformed $(x + 1)$ to standardize variances. $N = 3$.
NS > 0.05; [a] $P < 0.01$; [b] $P < 0.006$.

Table 3
Effects of Experiment (Experiment 1: May vs. Experiment 2: October 1995), habitat type (artificial seagrass versus oyster shell), and patch size (0.25, 0.5, 1, 4 m^{-2}) on the mean density of *Palaemonetes pugio*

Three-way ANOVA (model I).

Source of variation	*df*	MS	*F*
Experiment (A)	1	0.29	5.50[a]
Habitat type (B)	1	0.10	1.90 NS
Patch size (C)	3	0.45	8.52[b]
A × B	1	0.08	1.43 NS
A × C	3	0.09	1.70 NS
B × C	3	0.04	0.75 NS
A × B × C	3	0.01	0.26 NS
Error	30	0.05	

Shrimp density was log-transformed $(x + 1)$ to standardize variances. $N = 3$.
NS > 0.05; [a] $P < 0.02$; [b] $P < 0.001$.

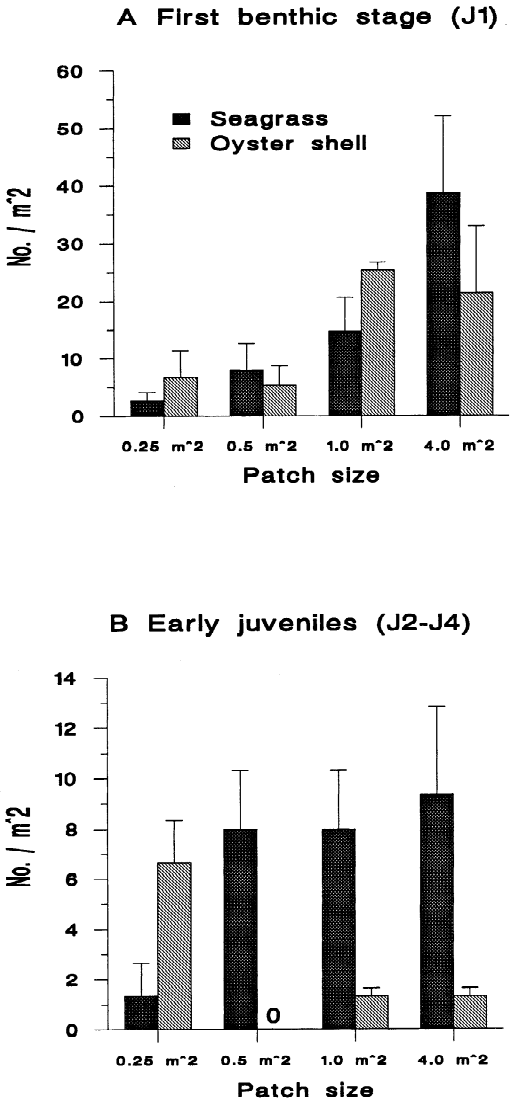

Fig. 4. Mean density (+ 1 SE) of (A) first benthic instar (J1) and (B) early juvenile (J2–J4) blue crabs (*Callinectes sapidus*) in four patch sizes of artificial seagrass and oyster shell-filled trays placed on unstructured bottom in October, 1995. Note different scales on *y*-axis. $N = 3$. See Section 3.2 for significance levels.

sapidus (J2–J4) varied according to Habitat type but not Patch size (two-way ANOVA; Substrate type: $F = 8.80$, $df = 1{,}16$, $P = 0.009$; Patch size: $F = 0.35$, $df = 3{,}16$, $P = 0.79$). The interaction effect was not significant ($F = 2.57$, $df = 3{,}16$, $P = 0.09$). There were significantly higher densities of J2–J4 stages of *C. sapidus* in seagrass than in oyster shell (lower-level ANOVA; Fig. 4b).

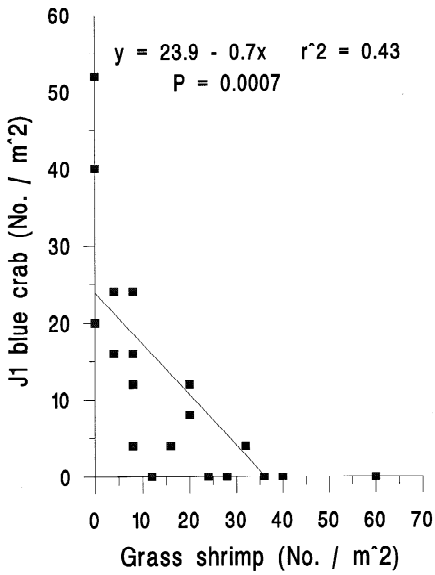

Fig. 5. Relationship between density of grass shrimp predators (>5.1 mm CL) and density of first benthic instar blue crab prey from artificial seagrass and oyster shell-filled plots placed on an unstructured bottom for 14 days in October, 1995. This figure represents 24 individual data points (2 Habitat types $\times$ 4 Patch sizes $\times$ 3 Replicates); however, many of these have the same x and y values, which mask numerous points.

3.3. Relationship between potential predators and early juvenile blue crab prey

There was a negative and significant relationship between the density of grass shrimp predators (*Palaemonetes intermedius*, *P. vulgaris* and *P. pugio* combined; >5.1 mm CL) and J1 prey (Fig. 5). Conversely, there was no significant relationship between the density grass shrimp predators and later stages (J2–J4) of blue crab prey (linear least-squares regression: $F = 0.23$, $df = 1{,}23$, $P = 0.64$). Moreover, there was no significant relationship between the density of potential fish predators (*Gobionellus bolesoma*, *G. shuffeldti*, *Lagodon rhomboides*, *Opsanus tau*) and J1 or J2–J4 prey (linear least-squares regression; all $P > 0.51$).

4. Discussion

A patch is often defined as a surface differing from its surrounding in terms of size, shape, habitat type and heterogeneity, and boundary characteristics (Weins, 1976; Forman and Godron, 1986). Patches can also be defined relative to the organism being studied and the questions being asked (e.g., Weins, 1976; Pickett and White, 1985; Addicott et al., 1987). Ideally, ‘patches’ should be defined using an objective criteria to define their boundaries (e.g., relative rate of change in a variable of interest per unit of

space; sensu Kotliar and Weins, 1990), and whether or not a particular organism or suite of organisms perceives it as a patch, or rather part of a contiguous habitat. The results from this study indicated that, regardless of habitat type, both grass shrimp and J1 blue crab respond to habitat patchiness at scales as small as 0.25 to 4 m^{-2}, whereas larger J2–J4 blue crab did not. Grass shrimp appeared to functionally perceive patch sizes between 0.25 and 0.5 m^{-2} as heterogeneous and responded with significantly higher abundances in the smallest patch (0.25 m^{-2}) compared to larger patches (0.5 to 4 m^{-2}). This pattern in abundance with patch size was consistent between three species of grass shrimp. Conversely, J1 blue crabs appeared to functionally perceive patch sizes between 1 and 4 m^{-2} as different and responded with significantly higher densities in the largest patch size (4 m^{-2}) compared to smaller patch sizes (0.25 to 1 m^{-2}). Thus, at relatively small spatial scales, altering the spatial scale of observation may alter grass shrimp and J1 blue crab abundance patterns, probably due to scale-dependent ecological processes underlying these patterns (e.g., Gascon and Travis, 1992; Levin, 1992). Determination of the 'grain' and 'extent' (sensu Kotliar and Weins, 1990) of grass shrimp and early juvenile blue crabs awaits further examination of abundance patterns across a range of spatial scales. Nevertheless, the results from this study suggest that the 'behavioral grain' of grass shrimp and J1 blue crab lies somewhere within the range of patch sizes examined in this study.

The use of artificial seagrass- and shell-filled plastic trays placed on an unstructured bottom allowed us to examine the interactive effects of experiment, patch size and habitat type upon the colonization success of four species of decapod crustaceans, while simultaneously controlling for microhabitat characteristics that often co-vary with patch size in nature (e.g., Hart and Horwitz, 1991; Irlandi, 1997). For example, the use of artificial seagrass and shell-filled trays eliminated any confounding effects of inter-patch distances, patch size and shape, age, blade density, shell volume, shell morphology and degree of epiphytism on experimental habitats. However, the use of artificial seagrass and shell-filled trays as analogies for natural habitats must be interpreted cautiously. The short-term response of organisms to novel substrata placed in their habitat may differ from that exhibited under totally natural circumstances. Nevertheless, the patterns of grass shrimp density with patch size observed in this study appear to be biologically realistic since similar patterns have been observed in natural seagrass patches and at larger spatial scales (D. Eggleston, unpubl. data; M. Fonseca, pers. comm.; E. Irlandi, pers. comm.). The value of artificial seagrass and oyster shell-filled trays has been demonstrated in numerous studies (e.g., Barber et al., 1979; Bell et al., 1985, 1987, 1995; Virnstein and Curran, 1986; Eggleston, 1988; Sogard, 1989; Bell and Hicks, 1991; Eggleston and Armstrong, 1995; Irlandi, 1997).

We examined the response of grass shrimp and J1 blue crabs at a relatively small spatial scale (0.25 to 4 m^{-2}) compared to more extensive seagrass and oyster reef systems. Since we chose to use artificial seagrass and shell-filled trays to avoid confounding habitat characteristics with patch size, we were constrained logistically to smaller patch sizes. However, the size distribution of natural seagrass and oyster patches is sometimes skewed towards patches <4 m^{-2} (Wells, 1961; Olesen and Sand-Jensen, 1994; Irlandi, 1997; Eggleston, in press).

4.1. Patterns and possible mechanisms for grass shrimp

Grass shrimp (*Palaemonetes* sp.) are highly mobile species that migrate between subtidal seagrass beds and salt marshes as the latter habitats are inundated during flooding tides (e.g., Kneib and Wagner, 1994). Within seagrass beds, grass shrimp may exhibit 50% turnover rates of individuals within a 3-h period, resulting in short-term distribution and abundance patterns that can vary on a daily basis (Howard, 1985). These observations of high mobility are consistent with the findings from colonization studies which indicate that grass shrimp densities in seagrass and macroalgal clumps can reach an asymptote after only 5–7 day post-defaunation (Stoner and Lewis, 1985; Sogard, 1989). Thus, our 14-day experimental period and low tide sampling regime seemed appropriate in terms of maximizing the density of grass shrimp in our artificial habitat plots, since soak time was beyond that presumably required for densities to reach an asymptote, and the artificial habitat plots were not competing with nearby salt marsh habitats during low tides when marshes were exposed.

In this study, densities of grass shrimp did not scale up directly with area. Rather, small patches, with greater perimeter:area ratios, contained significantly higher densities of grass shrimp than the largest patch size; a pattern which was consistent over each of our two 14-day experiments. These results are also consistent with findings from other studies which indicate increased recruitment of marine organisms in relatively small versus large habitat patches (e.g., Keough, 1984; Sousa, 1984; Bell et al., 1987; Sogard, 1989; Worthington et al., 1992; McNeill and Fairweather, 1993; Thrush et al., 1996), as well as observations that densities of grass shrimp are generally higher along marsh edges than in interior areas (Peterson and Turner, 1994). Our findings support the notion that shrimp are responding to an edge effect either as refuge, foraging, or both. For example, based upon changing perimeter:area ratios with patch size, as shrimp move between seagrass patches, or between seagrass and other habitats (e.g., soft-bottom, salt marsh), they were 2–4 times more likely to encounter one of our smaller patch sizes (0.25, 0.5, 1 m^{-2}) than our largest patch (4 m^{-2}). Similarly, if grass shrimp are using the patch edge as a refuge from predators (e.g., Coen et al., 1981), and periodically foraging on benthic infauna in the surrounding soft-bottom habitat (Posey and Hines, 1991), then small patches would contain 2–4 times the amount of foraging edge as our largest patch size. The relative importance of a foraging edge as opposed to a refuge edge in explaining the patterns of shrimp density with patch size observed in this study remain to be determined experimentally. Several alternative hypotheses to explain this density pattern with patch size would include predation being higher on shrimp in large versus small patches, or predators perceiving large patches as foraging sites causing shrimp to shift to smaller patches.

Previous experiments suggest that the high densities of grass shrimp typically observed in seagrass beds are primarily a result of predator avoidance behaviour and reduced predation, rather than increased food availability (Coen et al., 1981). Moreover, the three dimensional nature of seagrass provides additional space within which shrimp can distribute themselves, thereby reducing inter- and intra-specific displacement from patches when densities are relatively high (e.g., Van Dolah, 1976; Coen et al., 1981).

Relatively high densities of grass shrimp have also been observed in oyster reefs (Wells, 1961; Coen et al., in press; Posey et al., in press), which prompted our examination of habitat-specific responses by grass shrimp to habitat patchiness. For example, Coen et al., in press found mean grass shrimp densities of 160 m^{-2} in natural oyster reefs in South Carolina, USA. In North Carolina, USA, Posey et al. (in press) found considerable variability in grass shrimp densities on intertidal oyster reefs in marsh creeks; mean densities ranged from 0 to 800 m^{-2}.

The results from our study indicate that during the purported late spring/early summer peak in abundance of *Palaemonetes sp*. in North Carolina, significantly higher densities of shrimp occurred in seagrass compared to oyster shell. However, when shrimp densities were low in the fall, there was no difference in density between habitats. We hypothesize that when shrimp abundances are high during the late spring/early summer, and assuming food is not limiting, oyster habitats are unable to support relatively high densities due to a combination of high predation-induced mortality and increased agonist-induced displacement in oyster shell compared to seagrass. Alternatively, grass shrimp may actively select to reside in seagrass and salt marshes rather than oyster shell (e.g., Bell and Westoby, 1986).

4.2. Patterns and potential mechanisms for early juvenile blue crabs

Based on the increased perimeter to area ratio of small versus large patches, and given that the timing of our second experiment coincided with the ingress of settlement stages of blue crab megalopae, we expected that patterns of early juvenile blue crab density would be consistent with those observed for grass shrimp with highest densities in small versus large patches. Thus, we were surprised when, regardless of habitat type, the density of J1 blue crab increased with increasing patch size, and was significantly higher in the largest patch size. This pattern was exactly opposite to that observed for grass shrimp, and suggests two possible mechanisms. First, given that flow is reduced as water moves from the edge to the interior of a vegetated patch (Fonseca et al., 1982), large patches with greater total area versus edge should contain larger areas of attenuated flow than smaller patches (e.g., Bell et al., 1995). If we assume that the density of J1 reflects settlement patterns of megalopae, then the high density of J1 blue crabs in large patches is consistent with patterns expected if megalopae accumulated in zones of attenuated water flow. However, given the relatively strong swimming speeds of blue crab megalopae (4.8 $cm\,s^{-1}$; Luckenbach and Orth, 1992) compared to other marine invertebrate larvae (e.g., Butman, 1987), we suggest that a second mechanism, predation by grass shrimp, may also help to explain the observed pattern of J1 density with patch size. We found a negative, and highly significant relationship between the density of grass shrimp and J1 blue crabs. Moreover, recent laboratory settlement experiments showed that megalopal settlement was reduced significantly in the presence of chemical cues emanating from grass shrimp (Welch et al., in press). Predation by grass shrimp has been shown to structure benthic communities in a variety of systems (Bell and Coull, 1978; Nelson, 1981; Smith and Coull, 1987; Posey and Hines, 1991). Given that grass shrimp prey on blue crab megalopae (Olmi and Lipcius, 1991), we hypothesize that high predation-induced mortality in small patches by shrimp and high megalopal settlement in

large patches due to flow attenuation, jointly explain the increased density of J1 in large as opposed to small patches observed in this study.

In contrast to the patterns we observed for J1 blue crabs, the density of J2–J4 stages did not vary with patch size in either habitat type. However, the mean density of crabs going from the J1 to J2–J4 stages, changed in a disproportionate manner with patch size with the greatest reductions observed in the largest patch sizes (i.e., 1 and 4 m^{-2}). For example, in the 4 m^{-2} patches, there was a 3-fold decrease in mean crab density going from the J1 to the J2–J4 stages in seagrass, whereas there was a 10-fold decrease in oyster shell. For the 1 m^{-2} patch size, there was little reduction in mean crab density from the J1 to the J2–J4 stages in seagrass, but a 10-fold decrease in oyster shell, similar to that observed in the 4 m^{-2} patches. Thus, for similarly sized patches, reduction in mean crab density over time appears to be higher in oyster shell than seagrass. The mean density of J1 and J2–J4 blue crabs was nearly equal in the smaller patches (0.25 and 0.5 m^{-2}; Fig. 4). The greater reduction in mean crab densities in oyster shell versus seagrass, combined with proportionally greater reductions in mean crab density at high crab densities (i.e., large patch sizes), is consistent with the notion that early juvenile stages of *C. sapidus* exhibit density-dependent losses due to predation and emigration, with significantly higher loss rates in oyster shell than in seagrass. Recent findings from field and laboratory experiments in the Chesapeake Bay, USA indicate that J1–J4 stages of the blue crab exhibit density-dependent losses from seagrass habitats due to predation, cannibalism, and agonist-induced emigration (Pile et al., 1996; Moksnes et al., 1997).

The majority of research on the ecology of postlarval (megalopal) and early juvenile stages of the blue crab has focused on settlement, survival and movements in seagrass versus unvegetated habitats (Orth and van Montfrans, 1987; Williams et al., 1990; Heck and Coen, 1995; Morgan et al., 1996; Pile et al., 1996; and references therein). Yet, oyster reefs represent an alternative, structurally complex benthic habitat in most estuaries along the mid-Atlantic and Gulf coasts of the USA. In fact, oyster reefs are one of the only complex benthic habitats, along with salt marshes, present in estuaries along the southeastern coast of the USA (southern North Carolina through Georgia). Recently, the role of alternative habitats such as oyster reefs and shallow, coarse, woody debris as a refuge for grass shrimp and blue crabs has been identified (Ruiz et al., 1993; Dittel et al., 1995; Hines and Ruiz, 1995; Coen et al., in press; Posey et al., in press). For example, Posey et al. (in press) found higher densities of grass shrimp and blue crabs in subtidal oyster reefs than in inundated salt marshes. Moreover, their laboratory habitat choice experiments indicated that in the presence of fish predators, grass shrimp actively chose oyster shell over low density seagrass (Posey et al., in press). The results from this study and others (e.g., Posey et al., in press) suggest that blue crabs may rely equally on seagrass and oyster reefs as initial settlement and refuge habitat for early juveniles. Recent findings from laboratory settlement experiments suggest that blue crab megalopae are probably responding to the physical structure of oyster shells during settlement rather than chemical cues, since megalopae showed no settlement response to fouled oyster shells and avoided live oysters (Welch et al., in press).

Our study provides some of the first evidence that an organism's response to the spatial arrangement of habitats in a landscape, independent of the structural characteristics of the habitat, is species-specific. The results from this study suggest that the high

variability in grass shrimp and early juvenile blue crab abundance patterns often observed in field studies (e.g., Wells, 1961; Heck and Thoman, 1984; Wilson et al., 1990; Pile et al., 1996; Perkins et al., 1996; Posey et al., in press) may be due, in part, to the patch size of habitats that were sampled. We hypothesize that changes in patch size indirectly affect recruitment success of local populations of grass shrimp and blue crabs by altering foraging behaviour, predator distribution and abundance, and modifications of water flow. The strong edge effects observed for grass shrimp suggest that small patches may enhance the transfer of secondary production of infauna surrounding a patch to higher trophic levels (e.g., Welsh, 1975; Irlandi et al., 1995). The organism–structure relationships identified in this study using artificial seagrass and shell-filled trays, and conducted at relatively small scales of space (0.25 to 4 m^{-2}) and time (14 days), awaits further examination in natural beds and at larger scales. Nevertheless, the patterns observed in this study suggest numerous, explicit hypotheses and further highlight the importance of scale-dependent responses by mobile organisms to complex habitats.

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